IB Economics - Topic 2.9: Teacher and AI Grading Guide

Market Failure: Public Goods | Questions 1-20

Scope and assessment status

Each original paired question has (a) 10 marks and (b) 15 marks. All questions assess shared SL/HL content. Difficulty: 4 Easy, 10 Medium, 6 Hard pairs, distributed through the bank. Level and difficulty are independent.

Current IB Topic 2.9 covers non-rivalry, non-excludability, the free-rider problem, government direct provision and contracting out to the private sector. It lists no compulsory diagram or calculation.

Uploaded older IB textbook Chapter 5 section 5.5 discusses public goods and provision, but its quasi-public examples cannot expand the present topic beyond current IB requirements.

These are original adapted Paper 1-style practice questions, not official IB examination or marking material. The question bank revisits syllabus mechanisms through distinct explanatory and evaluative tasks; the pairs are alternatives for practice, not separate official learning outcomes.

The current syllabus lists public-good characteristics, free riding, direct provision and contracting out. Twenty distinct pairs cover those mechanisms. The requested 50 would require repeated answers or material from other topics; this shorter bank follows the master prompt's no-padding rule.

Coverage blueprint

Characteristics and free riding: 1-8

Distinguishing nearby categories: 9-12

Direct provision: 13-16

Contracting out and comparison: 17-20

Marking principles

Use best-fit assessment for each part, not a fixed marks-per-keyword checklist. Credit accurate alternatives addressing the task. Do not require every example or qualification below.

Part (a): 10 marks

Practice bands: 0; 1-2 limited; 3-4 mainly descriptive; 5-6 partial explanation; 7-8 clear and accurate causal analysis; 9-10 fully developed explanation with any required diagram or calculation. These abbreviated bands are not official IB descriptors.

Part (b): 15 marks

Practice bands: 0; 1-3 very limited; 4-6 superficial; 7-9 developed but weakly balanced; 10-12 balanced analysis and evaluation; 13-15 sustained, applied evaluation and supported judgment. A named case alone is not developed application.

Diagram and AI grading

No compulsory diagram. Assess both characteristics separately and distinguish free-rider underprovision from overuse of a common-pool resource. Optional illustrations should be evaluated on their explanation.

Record separate marks for (a) and (b), band evidence, diagram or numerical features if relevant, economic misconceptions, quality of real-world application, and one actionable improvement per part. If a student drawing is unclear, flag the uncertainty rather than assuming correctness.

Question 1

Easy | SL/HL | 25 marks

Topic 2.9: Characteristics

(a) [10] Explain why a pure public good is both non-rivalrous and non-excludable.

(b) [15] Using real-world examples, discuss whether every government-provided service is a public good.

Part (a) - Indicative answer and marking guidance

Essential concepts. A pure public good is non-rivalrous and non-excludable. Non-excludability creates a free-rider incentive and likely underprovision by private firms.

Required reasoning. One person's use does not diminish another's and exclusion from benefit is infeasible; funding source does not define classification.

Expected diagram or calculation. No diagram is required in Topic 2.9; classify characteristics accurately.

Common misconceptions. Do not equate government funding with public-good status; do not call a rivalrous common-pool resource a pure public good. For this question, check specifically that the student distinguishes characteristics from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. One person's use does not diminish another's and exclusion from benefit is infeasible; funding source does not define classification.

Counterargument and conditions. Funding and production do not determine rivalry or excludability.

Application and real-world example. National defence and public warnings offer contextual examples.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 2

Medium | SL/HL | 25 marks

Topic 2.9: Characteristics

(a) [10] Explain why national defence is commonly treated as a public good.

(b) [15] Using real-world examples, evaluate whether the benefits of public goods can be restricted to those who pay.

Part (a) - Indicative answer and marking guidance

Essential concepts. A pure public good is non-rivalrous and non-excludable. Non-excludability creates a free-rider incentive and likely underprovision by private firms.

Required reasoning. Protecting one resident does not reduce broad protection for others, and non-payers benefit.

Expected diagram or calculation. No diagram is required in Topic 2.9; classify characteristics accurately.

Common misconceptions. Do not equate government funding with public-good status; do not call a rivalrous common-pool resource a pure public good. For this question, check specifically that the student distinguishes characteristics from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Protecting one resident does not reduce broad protection for others, and non-payers benefit.

Counterargument and conditions. Funding and production do not determine rivalry or excludability.

Application and real-world example. National defence and public warnings offer contextual examples.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 3

Hard | SL/HL | 25 marks

Topic 2.9: Characteristics

(a) [10] Explain the difference between non-rivalry and non-excludability.

(b) [15] Using real-world examples, discuss whether a service can possess one characteristic without the other.

Part (a) - Indicative answer and marking guidance

Essential concepts. A pure public good is non-rivalrous and non-excludable. Non-excludability creates a free-rider incentive and likely underprovision by private firms.

Required reasoning. Non-rival is no diminished availability; non-excludability is inability to prevent access.

Expected diagram or calculation. No diagram is required in Topic 2.9; classify characteristics accurately.

Common misconceptions. Do not equate government funding with public-good status; do not call a rivalrous common-pool resource a pure public good. For this question, check specifically that the student distinguishes characteristics from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Non-rival is no diminished availability; non-excludability is inability to prevent access.

Counterargument and conditions. Funding and production do not determine rivalry or excludability.

Application and real-world example. National defence and public warnings offer contextual examples.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 4

Medium | SL/HL | 25 marks

Topic 2.9: Characteristics

(a) [10] Explain why a privately produced good can still have public-good characteristics.

(b) [15] Using real-world examples, evaluate whether public-good status requires government production.

Part (a) - Indicative answer and marking guidance

Essential concepts. A pure public good is non-rivalrous and non-excludable. Non-excludability creates a free-rider incentive and likely underprovision by private firms.

Required reasoning. Classification is about use and exclusion, independent of producer ownership.

Expected diagram or calculation. No diagram is required in Topic 2.9; classify characteristics accurately.

Common misconceptions. Do not equate government funding with public-good status; do not call a rivalrous common-pool resource a pure public good. For this question, check specifically that the student distinguishes characteristics from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Classification is about use and exclusion, independent of producer ownership.

Counterargument and conditions. Funding and production do not determine rivalry or excludability.

Application and real-world example. National defence and public warnings offer contextual examples.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 5

Medium | SL/HL | 25 marks

Topic 2.9: Free rider

(a) [10] Explain how non-excludability creates a free-rider problem.

(b) [15] Using real-world examples, discuss whether voluntary contributions can finance an adequate quantity of a public good.

Part (a) - Indicative answer and marking guidance

Essential concepts. A pure public good is non-rivalrous and non-excludable. Non-excludability creates a free-rider incentive and likely underprovision by private firms.

Required reasoning. Individuals may benefit without paying and have incentive to understate willingness to contribute.

Expected diagram or calculation. No compulsory diagram; show incentive to benefit without payment.

Common misconceptions. Do not equate government funding with public-good status; do not call a rivalrous common-pool resource a pure public good. For this question, check specifically that the student distinguishes free rider from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Individuals may benefit without paying and have incentive to understate willingness to contribute.

Counterargument and conditions. Some provision may arise from philanthropy, but adequate voluntary funding is uncertain.

Application and real-world example. National defence and warning-system funding are examples.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 6

Easy | SL/HL | 25 marks

Topic 2.9: Free rider

(a) [10] Explain why private firms may not supply a non-excludable good even when consumers value it.

(b) [15] Using real-world examples, evaluate whether an absence of paying customers means a good has no value.

Part (a) - Indicative answer and marking guidance

Essential concepts. A pure public good is non-rivalrous and non-excludable. Non-excludability creates a free-rider incentive and likely underprovision by private firms.

Required reasoning. Beneficiaries can avoid payment; firm cannot recover cost through ordinary user prices.

Expected diagram or calculation. No compulsory diagram; show incentive to benefit without payment.

Common misconceptions. Do not equate government funding with public-good status; do not call a rivalrous common-pool resource a pure public good. For this question, check specifically that the student distinguishes free rider from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Beneficiaries can avoid payment; firm cannot recover cost through ordinary user prices.

Counterargument and conditions. Some provision may arise from philanthropy, but adequate voluntary funding is uncertain.

Application and real-world example. National defence and warning-system funding are examples.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 7

Hard | SL/HL | 25 marks

Topic 2.9: Free rider

(a) [10] Explain why free riding can cause underprovision rather than zero provision.

(b) [15] Using real-world examples, discuss whether charities can completely resolve public-good underprovision.

Part (a) - Indicative answer and marking guidance

Essential concepts. A pure public good is non-rivalrous and non-excludable. Non-excludability creates a free-rider incentive and likely underprovision by private firms.

Required reasoning. Some contributors or patrons may fund partial provision, though incentives to free ride persist.

Expected diagram or calculation. No compulsory diagram; show incentive to benefit without payment.

Common misconceptions. Do not equate government funding with public-good status; do not call a rivalrous common-pool resource a pure public good. For this question, check specifically that the student distinguishes free rider from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Some contributors or patrons may fund partial provision, though incentives to free ride persist.

Counterargument and conditions. Some provision may arise from philanthropy, but adequate voluntary funding is uncertain.

Application and real-world example. National defence and warning-system funding are examples.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 8

Medium | SL/HL | 25 marks

Topic 2.9: Free rider

(a) [10] Explain why stated willingness to pay for a public good may differ from actual benefit received.

(b) [15] Using real-world examples, evaluate whether a survey alone is sufficient to choose the level of public provision.

Part (a) - Indicative answer and marking guidance

Essential concepts. A pure public good is non-rivalrous and non-excludable. Non-excludability creates a free-rider incentive and likely underprovision by private firms.

Required reasoning. People may misstate value if response influences payment or provision.

Expected diagram or calculation. No compulsory diagram; show incentive to benefit without payment.

Common misconceptions. Do not equate government funding with public-good status; do not call a rivalrous common-pool resource a pure public good. For this question, check specifically that the student distinguishes free rider from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. People may misstate value if response influences payment or provision.

Counterargument and conditions. Some provision may arise from philanthropy, but adequate voluntary funding is uncertain.

Application and real-world example. National defence and warning-system funding are examples.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 9

Hard | SL/HL | 25 marks

Topic 2.9: Classification

(a) [10] Explain why a free flood-warning message may have public-good aspects even if access requires registration.

(b) [15] Using real-world examples, discuss whether a registration requirement makes warning information a private good.

Part (a) - Indicative answer and marking guidance

Essential concepts. A pure public good is non-rivalrous and non-excludable. Non-excludability creates a free-rider incentive and likely underprovision by private firms. Government can directly provide or contract out production while funding access.

Required reasoning. Information can reach many at low marginal cost, but registration/coverage may make exclusion feasible.

Expected diagram or calculation. No compulsory diagram; assess rivalry and excludability separately, not price alone.

Common misconceptions. Do not equate government funding with public-good status; do not call a rivalrous common-pool resource a pure public good. For this question, check specifically that the student distinguishes classification from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Information can reach many at low marginal cost, but registration/coverage may make exclusion feasible.

Counterargument and conditions. Congestion and coverage can alter classification.

Application and real-world example. England Environment Agency free flood warnings with sign-up.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 10

Medium | SL/HL | 25 marks

Topic 2.9: Classification

(a) [10] Explain why a congested road is not a pure public good.

(b) [15] Using real-world examples, evaluate whether a road becomes a different type of good when tolls are introduced.

Part (a) - Indicative answer and marking guidance

Essential concepts. A pure public good is non-rivalrous and non-excludable. Non-excludability creates a free-rider incentive and likely underprovision by private firms. Government can directly provide or contract out production while funding access.

Required reasoning. One vehicle reduces available space when congested; toll permits exclusion.

Expected diagram or calculation. No compulsory diagram; compare both characteristics.

Common misconceptions. Do not equate government funding with public-good status; do not call a rivalrous common-pool resource a pure public good. For this question, check specifically that the student distinguishes classification from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. One vehicle reduces available space when congested; toll permits exclusion.

Counterargument and conditions. Rivalry depends on congestion, exclusion on access control.

Application and real-world example. A named toll road versus an untolled road.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 11

Easy | SL/HL | 25 marks

Topic 2.9: Classification

(a) [10] Explain why vaccination is not a pure public good even when it has wider social benefits.

(b) [15] Using real-world examples, discuss whether a positive externality is the same as a public-good characteristic.

Part (a) - Indicative answer and marking guidance

Essential concepts. A pure public good is non-rivalrous and non-excludable. Non-excludability creates a free-rider incentive and likely underprovision by private firms. Government can directly provide or contract out production while funding access.

Required reasoning. Each vaccine dose is used by one person and can be withheld; spillovers are separate.

Expected diagram or calculation. No compulsory diagram; do not conflate positive externality with non-rivalry.

Common misconceptions. Do not equate government funding with public-good status; do not call a rivalrous common-pool resource a pure public good. For this question, check specifically that the student distinguishes classification from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Each vaccine dose is used by one person and can be withheld; spillovers are separate.

Counterargument and conditions. Public funding does not erase rivalry/exclusion.

Application and real-world example. A national immunisation service.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 12

Medium | SL/HL | 25 marks

Topic 2.9: Classification

(a) [10] Explain why a fish stock differs from a pure public good even when exclusion is difficult.

(b) [15] Using real-world examples, evaluate whether all non-excludable resources face the free-rider problem in the same form.

Part (a) - Indicative answer and marking guidance

Essential concepts. A pure public good is non-rivalrous and non-excludable. Non-excludability creates a free-rider incentive and likely underprovision by private firms. Government can directly provide or contract out production while funding access.

Required reasoning. A catch reduces stock, so common-pool fish are rivalrous, unlike a pure public good.

Expected diagram or calculation. No compulsory diagram; distinguish rivalry and problem type.

Common misconceptions. Do not equate government funding with public-good status; do not call a rivalrous common-pool resource a pure public good. For this question, check specifically that the student distinguishes classification from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. A catch reduces stock, so common-pool fish are rivalrous, unlike a pure public good.

Counterargument and conditions. Management problem is overuse, not solely underprovision.

Application and real-world example. FAO fisheries versus national defence.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 13

Hard | SL/HL | 25 marks

Topic 2.9: Direct provision

(a) [10] Explain how direct government provision can overcome the private incentive to free ride.

(b) [15] Using real-world examples, discuss whether public finance guarantees the right quantity of a public good.

Part (a) - Indicative answer and marking guidance

Essential concepts. A pure public good is non-rivalrous and non-excludable. Non-excludability creates a free-rider incentive and likely underprovision by private firms.

Required reasoning. Tax funding spreads cost across beneficiaries without voluntary individual payment.

Expected diagram or calculation. No compulsory diagram; focus on free-riding and collective finance.

Common misconceptions. Do not equate government funding with public-good status; do not call a rivalrous common-pool resource a pure public good. For this question, check specifically that the student distinguishes direct provision from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Tax funding spreads cost across beneficiaries without voluntary individual payment.

Counterargument and conditions. Government still needs information about value and cost.

Application and real-world example. England flood-warning service and national defence.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 14

Medium | SL/HL | 25 marks

Topic 2.9: Direct provision

(a) [10] Explain why deciding how much of a public good to provide is difficult.

(b) [15] Using real-world examples, evaluate whether providing more warning coverage is always worth the additional cost.

Part (a) - Indicative answer and marking guidance

Essential concepts. A pure public good is non-rivalrous and non-excludable. Non-excludability creates a free-rider incentive and likely underprovision by private firms.

Required reasoning. Nonmarket value is hard to elicit, while extra coverage uses resources.

Expected diagram or calculation. No compulsory diagram or cost-benefit calculation required.

Common misconceptions. Do not equate government funding with public-good status; do not call a rivalrous common-pool resource a pure public good. For this question, check specifically that the student distinguishes direct provision from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Nonmarket value is hard to elicit, while extra coverage uses resources.

Counterargument and conditions. Marginal benefits may exceed or fall below costs depending on location.

Application and real-world example. England flood warnings with geographic limits.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 15

Medium | SL/HL | 25 marks

Topic 2.9: Direct provision

(a) [10] Explain why the financing of a free-at-use public good involves opportunity cost.

(b) [15] Using real-world examples, discuss whether a public good should be provided regardless of its cost.

Part (a) - Indicative answer and marking guidance

Essential concepts. A pure public good is non-rivalrous and non-excludable. Non-excludability creates a free-rider incentive and likely underprovision by private firms.

Required reasoning. Personnel and equipment have other uses even when users pay nothing.

Expected diagram or calculation. No compulsory diagram; separate user price from resource cost.

Common misconceptions. Do not equate government funding with public-good status; do not call a rivalrous common-pool resource a pure public good. For this question, check specifically that the student distinguishes direct provision from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Personnel and equipment have other uses even when users pay nothing.

Counterargument and conditions. High benefit may justify spending, but priorities compete.

Application and real-world example. Weather warning systems and national defence.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 16

Easy | SL/HL | 25 marks

Topic 2.9: Direct provision

(a) [10] Explain why government provision need not imply government production.

(b) [15] Using real-world examples, evaluate whether public agencies should perform every task in providing a public good.

Part (a) - Indicative answer and marking guidance

Essential concepts. A pure public good is non-rivalrous and non-excludable. Non-excludability creates a free-rider incentive and likely underprovision by private firms.

Required reasoning. Government can fund and oversee while another organisation produces.

Expected diagram or calculation. No compulsory diagram; distinguish funding, provision and production.

Common misconceptions. Do not equate government funding with public-good status; do not call a rivalrous common-pool resource a pure public good. For this question, check specifically that the student distinguishes direct provision from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Government can fund and oversee while another organisation produces.

Counterargument and conditions. Contract specification and monitoring affect outcomes.

Application and real-world example. UK Sourcing Playbook framework for insourcing/outsourcing.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 17

Hard | SL/HL | 25 marks

Topic 2.9: Contracting

(a) [10] Explain how contracting out production differs from direct production of a publicly financed good.

(b) [15] Using real-world examples, discuss whether private contractors can efficiently produce public goods.

Part (a) - Indicative answer and marking guidance

Essential concepts. A pure public good is non-rivalrous and non-excludable. Non-excludability creates a free-rider incentive and likely underprovision by private firms. Government can directly provide or contract out production while funding access.

Required reasoning. Government purchases output from contractor and retains financing/access decisions.

Expected diagram or calculation. No compulsory diagram; government remains responsible for provision.

Common misconceptions. Do not equate government funding with public-good status; do not call a rivalrous common-pool resource a pure public good. For this question, check specifically that the student distinguishes contracting from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Government purchases output from contractor and retains financing/access decisions.

Counterargument and conditions. Quality monitoring and tender competition condition performance.

Application and real-world example. UK Sourcing Playbook; hypothetical warning-system contractor.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 18

Medium | SL/HL | 25 marks

Topic 2.9: Contracting

(a) [10] Explain why contract design matters when a private firm delivers a public good on behalf of government.

(b) [15] Using real-world examples, evaluate whether the lowest bid guarantees the best public-good provision.

Part (a) - Indicative answer and marking guidance

Essential concepts. A pure public good is non-rivalrous and non-excludable. Non-excludability creates a free-rider incentive and likely underprovision by private firms. Government can directly provide or contract out production while funding access.

Required reasoning. Quality, coverage and accountability must be specified when users do not pay directly.

Expected diagram or calculation. No compulsory diagram; avoid equating a contract with market purchase by each beneficiary.

Common misconceptions. Do not equate government funding with public-good status; do not call a rivalrous common-pool resource a pure public good. For this question, check specifically that the student distinguishes contracting from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Quality, coverage and accountability must be specified when users do not pay directly.

Counterargument and conditions. Oversight costs can offset price savings.

Application and real-world example. Public procurement guidance in UK Sourcing Playbook.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 19

Hard | SL/HL | 25 marks

Topic 2.9: Contracting

(a) [10] Explain how imperfect information about service quality can affect a government decision to contract out.

(b) [15] Using real-world examples, discuss whether outsourcing is preferable when service quality is difficult to observe.

Part (a) - Indicative answer and marking guidance

Essential concepts. A pure public good is non-rivalrous and non-excludable. Non-excludability creates a free-rider incentive and likely underprovision by private firms. Government can directly provide or contract out production while funding access.

Required reasoning. Contractor may have incentive to meet observable targets while neglecting hard-to-measure quality.

Expected diagram or calculation. No compulsory asymmetric-information theory required beyond observability of quality.

Common misconceptions. Do not equate government funding with public-good status; do not call a rivalrous common-pool resource a pure public good. For this question, check specifically that the student distinguishes contracting from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Contractor may have incentive to meet observable targets while neglecting hard-to-measure quality.

Counterargument and conditions. Public production can also face performance problems.

Application and real-world example. Warning-system maintenance contract as scenario; UK sourcing framework.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Question 20

Medium | SL/HL | 25 marks

Topic 2.9: Comparison

(a) [10] Explain why direct public production and contracting out can both address the free-rider problem.

(b) [15] Using real-world examples, evaluate which approach is better for a public warning service.

Part (a) - Indicative answer and marking guidance

Essential concepts. A pure public good is non-rivalrous and non-excludable. Non-excludability creates a free-rider incentive and likely underprovision by private firms. Government can directly provide or contract out production while funding access.

Required reasoning. Both use collective funding; production method and accountability differ.

Expected diagram or calculation. No compulsory diagram; compare funding and service outcomes under each.

Common misconceptions. Do not equate government funding with public-good status; do not call a rivalrous common-pool resource a pure public good. For this question, check specifically that the student distinguishes comparison from nearby concepts and does not reverse the stated causal relationship.

Acceptable alternatives. Credit any other accurate economic argument, diagram or real-world example that directly addresses the specific question. Do not require a single policy conclusion or named case.

Part (b) - Indicative answer and marking guidance

Main argument. Both use collective funding; production method and accountability differ.

Counterargument and conditions. Capability, innovation, monitoring costs and access determine suitability.

Application and real-world example. England's free flood-warning service and UK sourcing guidance.

Evaluation and judgment. A strong response applies the example to the stated mechanism, weighs the qualification against the main argument, and reaches a conditional conclusion. Its conclusion may differ from the suggested route if the economics and case are sound.

Sources and scope

The current IB Economics guide controls syllabus boundaries. The uploaded Economics for the IB Diploma, second edition (2012), is a supplementary conceptual check; its older chapter sequence and content cannot expand the current specification. Real-world examples are optional and should be checked against their cited sources.

G. IB Economics guide, first assessment 2022, Topic 2.9, printed p. 31.

Open source

A. UK Environment Agency, free flood-warning service and geographic coverage.

Open source

B. UK Cabinet Office, Sourcing Playbook on insourcing and outsourcing public services.

Open source

Final checks

20 paired questions; 4 Easy, 10 Medium, 6 Hard; All questions assess shared SL/HL content. The student paper contains no answers or difficulty labels.